

Flight Radar – Operation

The device shows aircraft in the surrounding area. The round display is the radar; the small display shows details of the selected aircraft. Power on: connect a USB power supply. The radar appears after about 30 seconds.

Controls (rotary encoder)

Action	Effect
Turn (display shows ZOOM)	Change range: 10, 25, 50, 100 or 200 km
Short press	Switch between ZOOM and SELECT (shown at the bottom of the radar)
Turn (display shows SELECT)	Select an aircraft. The selected aircraft is marked red and its data appears on the small display.
Long press (about 1 second)	Next page on the small display (1/3, 2/3, 3/3, then back to 1/3)
No input for 10 seconds (in SELECT)	Returns to ZOOM automatically; the nearest aircraft is selected

Procedure: selecting an aircraft

1. Press the encoder briefly – the bottom of the radar shows **SELECT**.
 2. Turn the encoder until the desired aircraft is red.
 3. If needed, press and hold the encoder to see more data.
- The selection stays on the same aircraft between data updates.

Display

Radar (round display)	Meaning
Blue triangle	Own location (center)
Green dot	Aircraft
Red dot with ring	Selected aircraft
Rings / number at top	Distance; the number applies to the outer ring
N, E, S, W	North, east, south, west

Small display	Content
Page 1/3	Callsign, altitude (ft), speed (kt), heading, distance, aircraft type
Page 2/3	Operator, manufacturer and type, climb/descent rate (ft/min), ICAO address
Page 3/3	IP address, aircraft within range and within 200 km, nearest aircraft, age of data

Setup (first start, new WiFi or location)

1. On first start both displays show **SETUP**. To open setup later, hold the encoder pressed while powering on.
 2. Connect a phone or PC to the WiFi **FlightRadar-Setup** (no password). The setup page opens automatically, otherwise open **192.168.4.1**.
 3. Select the WiFi network and enter the password, latitude and longitude. OpenSky client ID and secret are optional.
 4. Tap **Save**. The device restarts and shows the radar.
- In the home network the page is also available at **flightradar.local** (or at the IP address on page 3/3).

Notes

- Data updates every 30 seconds, so the dots move in steps. “No traffic” means no aircraft within the radius: select a larger range.
- If the device does not respond: unplug USB, wait 10 seconds, plug in again. 1 ft = 0.30 m, 1 kt = 1.85 km/h. WiFi is required.

ESP32 Pinout

ESP32 DevKit (38 pin, AZ-Delivery). Run all components at 3.3 V. Disconnect the ESP32 from USB while wiring.

Component	Component pin	ESP32 pin	Done
Round display GC9A01 (SPI)	VCC	3V3	
	GND	GND	
	SCL (SCK)	GPIO18	
	SDA (MOSI)	GPIO23	
	CS	GPIO15	
	DC	GPIO2	
	RES	GPIO4	
OLED SSD1306 (I2C, 0x3C)	BLK	3V3	
	VCC	3V3	
	GND	GND	
	SDA	GPIO21	
Rotary encoder EC11	SCL	GPIO22	
	+	3V3	
	GND	GND	
	CLK	GPIO25	
	DT	GPIO26	
	SW	GPIO27	

Check by GPIO

GPIO	2	4	15	18	21	22	23	25	26	27
Function	DC	RES	CS	SCK	SDA OLED	SCL OLED	MOSI	CLK	DT	SW

Used: GPIO 2, 4, 15, 18, 21, 22, 23, 25, 26, 27. The round display and the OLED share no pins.

Firmware update (OTA)

1. Set an **OTA password** (at least 6 characters) on the setup page. Without a password, updates are disabled.
 2. Arduino IDE: select the network port **flightradar** under **Tools > Port**, upload and enter the OTA password.
 3. Or open **flightradar.local/update** in a browser (user **admin**, password = OTA password) and upload the .bin file (**Sketch > Export Compiled Binary**).
 4. The small display shows the progress, then the device restarts.
- The first OTA-capable firmware must be flashed over USB. If the sketch does not fit the update partition, select **Tools > Partition Scheme > Minimal SPIFFS (1.9MB APP with OTA)**.

Notes

- GPIO2 (DC) and GPIO15 (CS) are strapping pins. If the upload fails, briefly disconnect both wires.
- BLK is tied to 3V3 (GPIO21 is used for the OLED SDA).
- OLED stays dark and the monitor prints "SSD1306 init failed": check SDA/SCL, change 0x3C to 0x3D in the sketch.
- If one detent of the encoder skips two steps: in `handleEncoder()` change the threshold 4 to 2.
- WiFi, location, OpenSky credentials and OTA password are entered on the setup page and stored in flash. Nothing needs to be edited in the sketch. Serial Monitor: 115200 baud.